Chapter 11 Intro to Networking Exam

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Score: 98% Passing Score: 80%

Question 1: ✔ Correct

Which of the following WAN technologies provides digital dial-up connections on two 64 Kbps data channels?

- ➡ ISDN BRI
- ATM
- X.25
- Frame relay

EXPLANATION

ISDN BRI is a dial-up-only service. Basic Rate ISDN provides access to two 64 Kbps data channels (B channels) and one 16 Kbps service channel (D channel). The two B channels can be used together for a total data transfer rate of 128 Kbps. Data compression can further increase the data transfer rate.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 2: ✔ Correct

Your organization recently opened a branch office. You contracted with a WAN service provider to connect the branch office network to your home office network.

Recently, your CEO conducted a video conference with the employees at the branch office. The employees complained that the video was choppy and the audio was frequently out of sync with the video.

What is the most likely cause of this poor WAN performance?

- ➡ The WAN provider is throttling bandwidth on the link.
- There is an authentication misconfiguration issue between the WAN interfaces on both ends of the link.
- The WAN interface on the router at the branch office has been disabled.
- There is an IP address misconfiguration issue between the WAN interfaces on both ends of the link.
- There is a data link encapsulation protocol mismatch between the WAN interfaces on both ends of the link.

EXPLANATION

In this scenario, its possible that the WAN service provider is the cause of the problem. You should check the contract with the service provider to make sure they aren't throttling the bandwidth of the WAN link. It's not uncommon for service providers to impose bandwidth or utilization caps that could be hampering communications.

Because connectivity exists between the home and branch office networks in this scenario, the following are very unlikely to be the cause of the problem:

- A disabled WAN interface
- A protocol mismatch
- An authentication mismatch
- An IP address misconfiguration

REFERENCES

LabSim for Network Pro, Section 11.5.

Question 3: ✔ Correct

Which of the following technologies does GSM use to allow multiple connections on the same frequency?

- ➡ Time division multiple access
- Code division multiple access
- Multiple-input and multiple-output
- Frequency division multiple access

EXPLANATION

GSM uses time division multiple access (TDMA) to allow multiple connections on the same frequency.

Code division multiple access (CDMA) is not used by GSM, but it is the technology used by most mobile service providers in the United States. Multiple-input and multiple-output (MIMO) is a method for increasing data throughput and link range. Frequency division multiplexing (FDMA) is used primarily for satellite communications.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 4: ✔ Correct

Which of the following terms identifies the network of dial-up telephone and long-distance lines?

- Frame relay
- ATM
- VoIP
- PSTN
- ISDN

EXPLANATION

The Public Switched Telephone Network (PSTN) is the network used for placing local and long-distance phone calls. The local loop uses analog signals over POTS (regular telephone cable wires). The long-distance network typically uses digital signaling over fiber optic (typically using SONET).

Integrated Services Digital Network (ISDN) is a WAN technology that provides increased bandwidth within the local loop. Frame relay is a protocol used to connect to a WAN over dedicated (leased) lines. ATM is a WAN communication technology originally designed to carry time-sensitive data, such as voice and video. Voice over IP (VoIP) is a method for carrying phone calls over an IP-based network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 5: ✔ Correct

Which of the following ports are used with TACACS?

- 22
- 49
- 50 and 51
- 1812 and 1813
- 3389

EXPLANATION

Terminal Access Controller Access-Control System (TACACS) uses TCP and UDP ports 49.

Port 22 is used by Secure Shell (SSH). Ports 50 and 51 are used by IPsec. Ports 1812 and 1813 are used by Remote Authentication Dial-in User Service (RADIUS). Port 3389 is used by Remote Desktop Protocol (RDP).

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 6: ✔ Correct

Which network type divides transmitted data into smaller pieces and allows multiple communications on the network medium?

- Circuit-switched
- Sliding window
- ➡ Packet-switched
- Frame-based
- Multiplexed

EXPLANATION

A packet-switched network divides data into small units called packets. These packets are routed by their destination addresses. In a packet-switched network, multiple hosts can use the network medium at the same time. An Ethernet computer network is an example of a packet-switched network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 7: ✔ Correct

Which of the following protocols is used by PPP to enable support for multiple Network layer protocols?

- LCP
- SAP
- TCP
- HDLC
- ➡ NCP

EXPLANATION

PPP uses the network control protocol (NCP) to support multiple upper-layer protocols. LCP is used to establish the link, negotiate compression and authentication, detect errors, and tear down the link.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 8: ✔ Correct

When implementing a Multiprotocol Label Switching (MPLS) WAN, which data unit is managed by the routers at different sites?

- Bits
- Datagrams
- Frames
- ➡ Packets

EXPLANATION

In a Multiprotocol Label Switching (MPLS) WAN, IP packets are forwarded between sites. Packets are forwarded based on the contents of the label; the packet content is not examined. This allows for the creation of end-to-end circuits across any type of WAN transport medium using any protocol.

Bits are delivered with leased line WANs. Frames are delivered with frame relay and Ethernet WAN implementations.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 9: ✔ Correct

You are traveling throughout North America to many metropolitan and rural areas.

Which single form of internet connectivity provides the greatest potential connectivity wherever you travel?

- Broadband cable
- ISDN
- DSL
- ➡ PSTN

EXPLANATION

Network access using a modem over the telephone company network (PSTN) is not the fastest method for internet connectivity. However, it has the advantage of being available virtually anywhere that regular voice-grade communications are available.

Broadband cable is dependent on service offerings from the regional cable television company, which does not have as great a presence as the telephone company. To use broadband cable, the service must be added to the cable TV lines. DSL and ISDN are offered through the telephone company; however, they are not available in all service areas. And even when available, they require that the subscriber be within a certain proximity of telephone company equipment.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 10: ✔ Correct

Which WAN connection types use digital communications over public telephone lines? (Select two.)

- [ ] ATM
- [x] ISDN
- [ ] 56 Kbps dialup
- [ ] X.25
- [ ] SONET
- [x] DSL

EXPLANATION

Both DSL (Digital Subscriber Line) and ISDN (Integrated Services Digital Network) use regular phone lines for digital communications.

56 Kbps dialup uses analog signals over regular phone lines. X.25 can use either digital or analog signaling, but uses analog signaling when it is implemented over standard telephone lines. SONET operates over fiber optic cables. ATM uses fixed-length cells to transmit data over a variety of media (most commonly twisted pair and fiber optic).

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 11: ✔ Correct

Which of the following are characteristics of ATM? (Select two.)

- [ ] Supports variable-length packets.
- [x] Adds labels to data units.
- [ ] Uses POTS in the local loop.
- [x] Uses fixed-length cells of 53 bytes.
- [ ] Connects to the WAN cloud with a CSU/DSU.

EXPLANATION

ATM is a WAN communication technology originally designed for carrying time-sensitive data, such as voice and video. However, it can also be used for regular data transport.

- ATM is a packet switching technology that uses fixed-length data units called cells. Each cell is 53 bytes.
- The cell header includes labels that identify the virtual path information. ATM switches in the WAN cloud use the virtual path to switch cells within the WAN to the destination.

MPLS uses labels but supports variable-length packets. ISDN is a WAN technology that uses existing telephone lines (POTS) to connect to the WAN cloud. Both frame relay and ISDN use a CSU/DSU to connect the line to the WAN link.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 12: ✔ Correct

Which of the following are benefits of LCP? (Select three.)

- [x] Negotiates the use (or lack) of authentication before starting the session
- [ ] Provides physical addressing for the data link layer
- [x] Monitors data dropped on the link and avoids frame looping
- [ ] Provides logical addressing for the data link layer
- [x] Provides load balancing across multiple links

EXPLANATION

Benefits of LCP include the following:

- Negotiates the use of authentication before starting the session.
- Monitors data dropped on the link and avoids frame looping (error detection).
- Compresses data at the source and decompresses data at the destination (compression).
- Provides load balancing across multiple links (multilink).

LCP does not encapsulate protocols; NCP handles that task.

LCP is a data link layer protocol, so it does not provide support for physical interfaces. Because PPP is a point-to-point protocol, no logical or physical addressing is necessary.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 13: ✔ Correct

Which of the following is true of Multiprotocol Label Switching? (Select two.)

- [ ] Packets routed using MPLS require additional routing lookup overhead.
- [ ] MPLS-enabled routers switch packets based on the packet contents, not on label information.
- [x] CEF is required for all MPLS-enabled Cisco routers.
- [ ] It is designed to forward only packets created with the TCP/IP protocol.
- [x] It can carry ATM, SONET, and Ethernet traffic.

EXPLANATION

Key facts about Multiprotocol Label Switching (MPLS) include the following:

- It can carry ATM, SONET, and Ethernet traffic.
- CEF is required for all MPLS-enabled Cisco routers.
- Packets are routed with a minimal amount of routing lookup overhead.
- MPLS-enabled routers switch packets based on the label information, not on the packet contents.
- It is designed to support the forwarding of protocols other than TCP/IP.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 14: ✔ Correct

What is the maximum data rate of an ISDN BRI line?

- 64 Kbps
- 128 Kbps
- 1.544 Mbps
- 256 Kbps

EXPLANATION

IDSN BRI carries two B channels and one D channel. Each B channel can maintain a throughput of 64 Kbps for a total of 128 Kbps. The D channel is used for line management and does not contribute to data throughput.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 15: ✕ Incorrect

Which of the following are characteristics of TACACS+? (Select two.)

- [ ] Uses TCP.
- [x] Allows the possibility of three different servers, one each for authentication, authorization, and accounting.
- [ ] Allows the possibility of two different servers, one for authentication and authorization, and another for accounting.
- [x] ~~Uses UDP.~~

EXPLANATION

TACACS+ was originally developed by Cisco for centralized remote access administration. TACACS+:

- Provides three protocols, one each for authentication, authorization, and accounting. This allows each service to be provided by a different server.
- Uses TCP.
- Encrypts the entire packet contents.
- Supports more protocol suites than RADIUS.

RADIUS is used by Microsoft servers for centralized remote access administration. RADIUS:

- Combines authentication and authorization using policies to grant access.
- Uses UDP.
- Encrypts only the password.
- Often uses vendor-specific extensions. RADIUS solutions from different vendors might not be compatible.

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 16: ✔ Correct

Which of the following are characteristics of MPLS? (Select two.)

- [x] Adds labels to data units
- [ ] Uses fixed-length data units
- [x] Supports variable-length data units
- [ ] Uses a committed information rate (CIR)
- [ ] Is a local loop technology

EXPLANATION

MPLS is a WAN data classification and data carrying mechanism.

- MPLS is a packet switching technology that supports variable-length data units.
- MPLS adds a label to packets between the existing Network and Data Link layer formats. Labels are added when the packet enters the MPLS network and removed when the packet exits the network.
- Information in the label is used to switch the packet through the MPLS network to the destination.
- MPLS labels can identify the route or even the network type to use. MPLS labels are often used to provide different classes of service for data streams.

ATM uses labels, but uses fixed-length cells. ISDN is a local loop technology. Frame relay uses the committed information rate (CIR) to define levels of service. MPLS service levels are provided by information within the labels.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 17: ✔ Correct

This question includes an image to help you answer the question. View Image

Consider the network shown in the exhibit.

You have been experiencing intermittent connectivity issues with switch2. To check the status of the interfaces, you run the following commands:

```
switch2# show interfaces fa0/1 status
Port Name Status Vlan Duplex Speed Type
Fa0/1 connected 3 a-half a-100 10/100BaseTX
```

```
switch2# show interfaces Gi0/1 status
Port Name Status Vlan Duplex Speed Type
Gi0/1 connected trunk a-full a-1000 1000BaseTX
```

```
switch2# show interfaces Gi0/2 status
Port Name Status Vlan Duplex Speed Type
Gi0/2 connected trunk a-full a-1000 1000BaseTX
```

What is the issue with this network?

- There is a link speed mismatch on the Gi0/2 interface.
- The device connected to the Gi0/1 interface has auto-negotiation disabled.
- There is a link speed mismatch on the Gi0/1 interface.
- There is a duplex mismatch on the Gi0/2 interface.
- ➡ The device connected to the Fa0/1 interface has auto-negotiation disabled.

EXPLANATION

A duplex mismatch probably exists on the Fa0/1 interface. Note that duplexing has been automatically set to half, which is the default behavior for Cisco devices when auto-negotiation fails. To fix the issue, check the Gi0/1 interface on router1 to see if auto-negotiation has been disabled. You could manually configure the Fa0/1 interface on switch2 to use the same duplexing and link speed settings as the interface on the router, or you could re-enable auto-negotiation on the router interface.

The Gi0/1 and Gi0/2 interfaces on switch2 appear to be functioning correctly with full duplexing and full link speed automatically configured.

REFERENCES

LabSim for Network Pro, Section 11.5.

Question 18: ✔ Correct

Which of the following is a characteristic of TACACS+?

- ➡ Encrypts the entire packet, not just authentication packets.
- Uses UDP ports 1812 and 1813.
- Requires that authentication and authorization are combined in a single server.
- Supports only UDP.

EXPLANATION

TACACS+ was originally developed by Cisco for centralized remote access administration. TACACS+:

- Provides three protocols, one each for authentication, authorization, and accounting. This allows each service to be provided by a different server.
- Uses TCP port 49.
- Encrypts the entire packet contents, not just authentication packets.
- Supports more protocol suites than RADIUS.

RADIUS is used by Microsoft servers for centralized remote access administration. RADIUS:

- Combines authentication and authorization using policies to grant access.
- Allows for the separation of accounting to different servers. However, authentication and authorization remain combined on a single server.
- Uses UDP ports 1812 and 1813.
- Uses a challenge/response method for authentication. RADIUS encrypts only the password using MD5.

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 19: ✔ Correct

Two routers with the host names SLC and PROVO have been configured to connect using PPP with CHAP authentication through their BRI0 interfaces. Attempts to establish a session between the two routers fail. You check the running configuration on both routers and find the output shown below:

```
hostname SLC
enable password cisco
username PROVO password vanilla
!
!
interface Serial0
ip address 172.16.55.129
255.255.255.252
encapsulation ppp
ppp authentication chap
!
! - remaining output omitted --
```

```
hostname PROVO
enable password ccna
username SLC password chocolate
!
!
interface Serial0
ip address 172.16.55.130
255.255.255.252
encapsulation ppp
ppp authentication chap
!
! - remaining output omitted --
```

What should you do to correct the problem?

- On SLC, change the username to **SLC**. On PROVO, change the username to **PROVO**.
- On PROVO, change the enable password to **cisco**.
- On SLC, change the username password to **ccna**. On PROVO, change the username password to **cisco**.
- ➡ On SLC, change the username password to **chocolate**.
- On PROVO, change the IP address to 172.16.55.128.

EXPLANATION

The username passwords used by each router must match. In this scenario, changing the username password on SLC to **chocolate** would correct the problem (you could also change the password on PROVO to **vanilla**).

The username configured on each router must match the host name of the remote router that it will be connecting to. The IP addresses assigned to the interfaces are both on the 172.16.55.128 subnet. You cannot assign that address to a host.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 20: ✔ Correct

You have installed an ISDN connection into your home so you can connect to the internet and talk on the phone at the same time. While you are talking on the telephone, what is the maximum data rate of your internet connection?

- ➡ 64 Kbps
- 1.544 Mbps
- 128 Kbps
- 56.8 Kbps

EXPLANATION

Your ISDN line has two B channels. Each B channel transmits data at 64 Kbps, giving you a maximum of 128 Kbps. However, one of your B channels is being used when you use the phone; therefore, the maximum speed while the phone is in use is 64 Kbps.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 21: ✔ Correct

Which of the following technologies uses variable-length packets, adds labels to packets as they enter the WAN cloud, and uses the labels to switch packets and prioritize traffic?

- Frame relay
- SONET
- ➡ MPLS
- ISDN
- ATM

EXPLANATION

MPLS is a WAN data classification and data carrying mechanism.

- MPLS is a packet switching technology that supports variable-length frames.
- MPLS adds a label to packets between the existing Network and Data Link layer formats. Labels are added when the packet enters the MPLS network and removed when the packet exits the network.
- Information in the label is used to switch the packet through the MPLS network to the destination.
- MPLS labels can identify the route or even the network type to use. MPLS labels are often used to provide different classes of service for data streams.

ATM is a WAN communication technology that uses labels but has fixed-length cells of 53 bytes. Frame relay is a protocol used to connect to a WAN over dedicated (leased) lines. ISDN is a WAN technology that provides increased bandwidth within the local loop. SONET is a standard for networking over an optical medium. SONET is classified as a transport protocol, in that it can carry other types of traffic such as ATM, Ethernet, and IP. Most PSTN networks use SONET within the long-distance portion of the PSTN network.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 22: ✔ Correct

Which of the following are methods for providing centralized authentication, authorization, and accounting for remote access? (Select two.)

- [ ] PKI
- [ ] AAA
- [ ] 802.1x
- [ ] EAP
- [x] RADIUS
- [x] TACACS+

EXPLANATION

Both RADIUS and TACACS+ are protocols used for centralized authentication, authorization, and accounting used with remote access. Remote access clients send authentication credentials to remote access servers. Remote access servers are configured as clients to the RADIUS or TACACS+ servers and forward the authentication credentials to the servers. The servers maintain a database of users and policies that control access for multiple remote access servers.

AAA stands for authentication, authorization, and accounting, and is a generic term that describes the functions performed by RADIUS/TACACS+ servers. A Public Key Infrastructure (PKI) is a system of certificate authorities that issue certificates. 802.1x is an authentication mechanism for controlling port access. 802.1x uses RADIUS/TACACS+ servers. EAP is an authentication protocol that allows the use of customized authentication methods.

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 23: ✔ Correct

What must you install between your network and a T1 line for your network to use the T1 line?

- Transceiver
- Bridge
- Gateway
- ➡ CSU/DSU

EXPLANATION

A CSU/DSU is required for a network to send signals on a T1 line.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 24: ✔ Correct

You are moving to an area where DSL will be available in the next six months. Which method of internet connectivity should you implement until DSL is available if your existing connectivity needs are minimal?

- ➡ PSTN
- Cable modem
- ISDN
- Satellite

EXPLANATION

Dial-up networking using the public switched telephone network (PSTN) offers sufficient network connectivity for a relatively minimal investment. You can use dial-up with little hardware, setup, or connection costs.

The other modes of networking provide greater capability than you require at more of an investment in equipment than is worthwhile for such a short period of time.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 25: ✔ Correct

You are troubleshooting physical layer issues with the Gi0/1 interface in a router. You need to view and analyze the number of collisions detected on the interface.

Which command should you use?

- ➡ `show interfaces gi0/1`
- `show controllers gi0/1`
- `show interfaces gi0/1 description`
- `show interfaces gi0/1 status`

EXPLANATION

The **show interfaces gi0/1** command displays statistics about the Gi0/1 interface, including the number of collisions and late collisions.

The **show interfaces gi0/1 status** command displays summary information about the interface status. The output displays the port, name, status, VLAN assignment, duplex configuration, interface speed, and link type. The **show interfaces gi0/1 description** command displays the line and protocol status of the interface. The **show controllers** command is used to display configuration parameters for serial interfaces (not Ethernet interfaces), such as the type of serial cable and which end of the cable is connected to the device (DCE or DTE).

REFERENCES

LabSim for Network Pro, Section 11.5.

Question 26: ✔ Correct

Which of the following protocols can be used to centralize remote access authentication?

- ➡ TACACS
- CHAP
- Kerberos
- SESAME
- EAP

EXPLANATION

Centralized remote access authentication protocols include:

- Remote Authentication and Dial-In User Service (RADIUS)
- Terminal Access Controller Access Control System (TACACS)

Password authentication protocol (PAP) and challenge handshake authentication protocol (CHAP) are authentication protocols used between the client and the server. Kerberos and Secure European System for Applications in a Multi-Vendor Environment (SESAME) are single sign-on protocols.

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 27: ✔ Correct

Which of the following are the the WAN service provider's responsibility to maintain? (Select three.)

- [x] CO
- [x] Local loop
- [x] PSE
- [ ] CPE
- [ ] DTE

EXPLANATION

WAN service providers are responsible for:

- Packet switching exchange (PSE)—equipment inside the WAN cloud.
- Central office (CO)—equipment that allows access to the PSE.
- Local loop—wiring that connects the customer to the CO.

Customers are responsible for:

- Customer premises equipment (CPE)—any equipment at the customer site.
- Data terminal equipment (DTE)—devices that send local data to the WAN.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 28: ✔ Correct

PPP supports authentication, compression, and multiple Network layer protocols.

Which of the following correctly sequences these functions when a PPP link is established?

- Perform authentication, negotiate compression settings, negotiate Network layer protocols
- Negotiate Network layer protocols, perform authentication, negotiate compression settings
- Perform authentication, negotiate Network layer protocols, negotiate compression settings
- ➡ Negotiate compression settings, perform authentication, negotiate Network layer protocols
- Negotiate Network layer protocols, negotiate compression settings, perform authentication
- Negotiate compression settings, negotiate Network layer protocols, perform authentication

EXPLANATION

PPP uses the following process to open a session:

1. Exchange LCPs to establish the link and negotiate communication parameters (such as compression settings).
2. Perform authentication (optional).
3. Exchange NCPs to negotiate the Network layer protocols.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 29: ✔ Correct

Which of the following WAN technologies provides packet switching over high-quality digital lines at speeds greater than 1.544 Mbps?

- ➡ (●) Frame relay
- ( ) X.25
- ( ) POTS
- ( ) ISDN

EXPLANATION

Frame relay networks are packet-switched networks that operate at T1 or T3 speeds. Frame relay achieves faster speeds by using high-quality digital lines, which require less extensive error control than X.25 networks.

POTS stands for plain old telephone system. In the United States, ISDN networks can only offer speed up to 1.544 Mbps.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 30: ✔ Correct

Which of the following describe the channels and data transfer rates used for ISDN BRI? (Select two.)

- [ ] One D channel operating at 64 Kbps
- ➡ [x] Two B channels operating at 64 Kbps each
- [ ] 23 B channels operating at 64 Kbps each
- ➡ [x] One D channel operating at 16 Kbps

EXPLANATION

ISDN BRI uses two B channels operating at 64 Kbps each and one D channel operating at 16 Kbps. It is often called 2B + 1D.

ISDN PRI uses 23 B channels (at 64 Kbps) and one D channel (at 64 Kbps). It is also called 23B + 1D.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 31: ✔ Correct

Which type of internet service uses the DOCSIS specification?

- Unshielded twisted pair
- ➡ Coaxial cable
- Fiber optic
- Shielded twisted pair

EXPLANATION

The Data Over Cable Service Interface Specification (DOCSIS) defines coaxial cable networking specifications. It is used by cable TV providers to provide internet access over their existing coaxial cable infrastructure. It specifies channel widths and modulation techniques. It also defines the manner in which the core components of the network communicate.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 32: ✔ Correct

Which of the following internet connection technologies requires that the location be within a limited distance of the telephone company central office?

- ➡ DSL
- Satellite
- Cable modem
- Wireless

EXPLANATION

There are several variations of the digital subscriber line (DSL) technology, which are collectively referred to as xDSL. DSL operates over existing telephone company copper wires. DSL operates concurrently with regular voice-grade communications by utilizing higher frequencies unused by voice transmissions. One of the consequences of splitting the signal in this manner is that DSL must operate within a fixed distance of the telephone company's network switching equipment.

A cable modem can be provided as a means of internet access by the cable television company anywhere within the service area of the cable television company.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 33: ✔ Correct

Which of the following statements about the functionality of LCP are true? (Select three.)

- [ ] LCP provides remote terminal access in public data networks.
- [x] Data can be compressed at the source and decompressed at the destination.
- [ ] LCP encapsulates multiple protocols.
- [x] LCP provides multilink support.
- [ ] LCP is responsible for logical addressing in PPP.
- [x] Usernames and passwords may be required during the handshake.

EXPLANATION

The link control protocol (LCP) is a sublayer within the PPP protocol stack. LCP options provide authentication, compression, error detection, and multilink functionality.

Multiple protocols are encapsulated in PPP by NCP. LCP is not a Layer 3 protocol, so it does not provide logical addressing. Remote terminal access is a function of the X.25 protocol.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 34: ✔ Correct

Which three of the following are advantages of ISDN BRI?

- [x] It has faster connect times than traditional dial-up access.
- [ ] It supports data transfer rates up to 1.55 Mbps.
- [x] It uses existing telephone lines for digital data transmission.
- [ ] It offers a dedicated always-on connection between two points.
- [x] It can be used to transmit multiple data types on the same line (data, voice, video, and so on).

EXPLANATION

ISDN BRI is similar to a dial-up connection, but has faster connect times and data transfer rates than traditional dial-up using 56 Kbps modems. It uses existing 4-wire telephone lines in the local loop and can be used to simultaneously transmit multiple data types. Data transfer rates are up to 128 Kbps.

REFERENCES

LabSim for Network Pro, Section 11.1.

Question 35: ✔ Correct

This question includes an image to help you answer the question. **View Image**

Review the output from the `show interfaces fa0/1` command on the switch2 switch in the exhibit.

What is wrong with the fa0/1 interface in this example?

- A duplex mismatch exists with the device on the other end of the connection.
- The line status is administratively down.
- The protocol status is down.
- The cable connecting the fa0/1 interface with the device on the other end is a straight-through cable, but needs to be crossed-over.

EXPLANATION

In this example, the following statistics indicate that a duplex mismatch error has occurred:

- Duplexing is set to half.
- There are a significant number of runts.
- There are a significant number of collisions.
- There are a significant number of late collisions.

REFERENCES

LabSim for Network Pro, Section 11.5.

Question 36: ✔ Correct

Your client has acquired several small companies and would like to connect them together into one network. Not all of the routers are Cisco devices, and compatibility is a concern.

Which WAN encapsulation method should you recommend your client use?

- Cisco HDLC
- ➡ PPP
- Ethernet
- PAP

EXPLANATION

The point-to-point (PPP) protocol is not proprietary. For this reason, it is the best choice for connecting dissimilar vendor devices.

Cisco HDLC is the default serial encapsulation method, but it is only supported on Cisco devices. PAP is a PPP authentication protocol. Ethernet is not a WAN protocol.

REFERENCES

LabSim for Network Pro, Section 11.2.

Question 37: ✔ Correct

Which of the following describe the EDGE cellular technology? (Select two.)

- [ ] Uses MIMO
- [ ] An extension to 3G
- ➡ [x] Offers speeds of 400–1,000 Kbps
- ➡ [x] The first internet-compatible technology

EXPLANATION

The EDGE cellular technology was an intermediary between 2G and 3G networks. EDGE was the first cellular technology to be truly internet-compatible and has speeds of 400–1,000 Kbps.

MIMO is used by HSPA+, LTE, and 4G networks.

REFERENCES

LabSim for Network Pro, Section 11.3.

Question 38: ✔ Correct

You are reviewing the output of the **`show interfaces`** command for the Gi0/1 interface on a switch. You notice a significant number of CRC errors displayed.

What are the most likely causes? (Select two. Each response is a complete solution.)

- [ ] The device on the other end of the cable is powered off or the other interface is administratively shutdown.
- [ ] The cable connected to this interface is a straight-through cable, but should be a cross-over cable.
- [x] EMI or cross-talk on the cable connected to the interface.
- [ ] The cable connected to this interface is a cross-over cable, but should be a straight-through cable.
- [x] Collisions.

EXPLANATION

CRC errors are received frames that did not pass the FCS check. These are usually caused by collisions, but they can also be caused by EMI or cross-talk on UTP cabling. All of these conditions can damage frames on the wire, causing a CRC error.

Using the wrong type of cabling would cause the link to go down. A disabled interface on the other end of the cable would also cause the link to go down.

REFERENCES

LabSim for Network Pro, Section 11.5.

Question 39: ✔ Correct

You have just signed up for internet access using a local provider that gives you a fiber optic line into your house. From there, Ethernet and wireless connections are used to create a small network within your home.

Which of the following protocols would be used to provide authentication, authorization, and accounting for the internet connection?

- PPP
- ➡ PPPoE
- ICA
- RDP
- L2TP

EXPLANATION

PPP over Ethernet (PPPoE) is used for connections that have an always on state, such as DSL or fiber optic running Ethernet. PPPoE is a modification of PPP that allows for negotiation of additional parameters that are typically not present on a regular Ethernet network. ISPs typically implement PPPoE to control and monitor internet access over broadband links.

The point-to-point protocol (PPP) is used for dial-up connections. RDP and ICA are Remote Desktop protocols. L2TP is a VPN protocol.

REFERENCES

LabSim for Network Pro, Section 11.4.

Question 40: ✔ Correct

You often travel away from the office. While traveling, you would like to use a modem on your laptop computer to connect directly to a server in your office and access files on that server that you need.

You want the connection to be as secure as possible. Which type of connection will you need?

- Internet
- Intranet
- ➡ Remote access
- Virtual private network

EXPLANATION

Use a remote access connection to connect directly to a server at a remote location.

You could use a VPN connection through the internet to connect to the server security. However, the connection would involve connecting to the internet through a local ISP, then establishing a VPN connection to the server. While the VPN connection through the internet is secure, it is not as secure as a direct remote connection to the server.

An intranet is an internal network that only internal users can access.

REFERENCES

LabSim for Network Pro, Section 11.4.

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